

EXISTENCE OF SINGULAR OPTIMAL CONTROL LAWS FOR STOCHASTIC DIFFERENTIAL EQUATIONS*

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This paper studies the existence of a control problem where the state is governed by a stochastic differential equation allowing both classical control and singular control. By transforming it into a new problem which involves only classical control, it is shown that there exists an optimal control where the classical variable is in Markovian form and the increment of the singular control variable on any time interval is adapted to the state process on the same time interval.

KEY WORDS: singular controls, singular control laws, existence theory

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1. INTRODUCTION

The class of singular stochastic control problems, which has been studied extensively in recent years, deals with systems described by a stochastic differential equation in which one restricts the cumulative displacement of the state caused by control to be of an additive nature. More precisely, in this paper we study the existence of optimal controls to the problem in which the state evolves according to the d -dimensional stochastic differential equation

$$x(t) = x + \int_s^t b(\theta, x(\theta), u(\theta)) d\theta + \int_s^t \sigma(\theta, x(\theta), u(\theta)) dB_\theta + v_t^1 - v_t^2,$$

where $b(\cdot, \cdot, \cdot)$, $\sigma(\cdot, \cdot, \cdot)$ are given deterministic functions, $(B_t, t \geq 0)$ is a d -dimensional Brownian motion, x is the initial state at time s and u , v^1 , v^2 stand for the control variables with v^1 , v^2 nondecreasing componentwise.

The expected cost has the form

$$J(\alpha) \equiv E^P \left\{ \int_s^T f(t, x_t, u_t) dt + \int_{[s, T)} c(t) \cdot dv_t^1 + \int_{[s, T)} \bar{c}(t) \cdot dv_t^2 \right\},$$

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been unable to show that the optimal state process is a reflected diffusion in this region.

A similar approach has been used by Martins and Kushner [24], Kurtz [25], and Zhu [30] to study the weak convergence of probability distributions.

This paper is organized as follows: both the singular control and the control problems are introduced in Section 2. In Section 3 the equivalence of the two problems is established. Section 4 studies the existence of the optimal control, and some comments about the approach in this paper are given in Sections 5. For the reader's convenience we give some analysis results used in this paper in the Appendix.

2. FORMULATION OF THE PROBLEM

Throughout this paper, we assume U is a compact metric space, $T > 0$ is fixed, $\Sigma = [0, T] \times \mathbb{R}^d$. We denote by $\mathcal{S}^d$ the space of $d \times d$ -matrices, and $\mathcal{D}^d[0, T]$ the space of $\mathbb{R}^d$ -valued functions defined on $[0, T]$ that have left continuous and nondecreasing components. We assume the following functions are given,

- $\sigma: \Sigma \times U \mapsto \mathcal{S}^d$ and $b: \Sigma \times U \mapsto \mathbb{R}^d$ are bounded and continuous in (t, x, u) ,
- $f: \Sigma \times U \mapsto \mathbb{R}$ is a lower semicontinuous function in (t, x, u) and such that

$$-K \leq f(t, x, u) \leq C(1 + \|x\|^m), \quad (t, x, u) \in \Sigma \times U,$$

for some constants $m, K, C > 0$, and

- $c = (c^i)$, $\bar{c} = (\bar{c}^i): [0, T] \mapsto \mathbb{R}_+^d$ are bounded and lower semicontinuous functions on $[0, T]$, and $c^i, \bar{c}^i > 0$, $1 \leq i \leq d$.

Moreover we let $a = \sigma \sigma^T$ and we assume

- for $(t, x) \in \Sigma$,

$$K(t, x) \equiv \{(a(t, x, u), b(t, x, u), z): z \geq f(t, x, u), u \in U\} \subset \mathcal{S}^d \times \mathbb{R}^d \times \mathbb{R}$$

is convex.

The last condition, sometimes called the Roxin condition, allows us to replace singular controls by ordinary controls u_t . It is satisfied for example if a, b are linear in u and U is convex.

2.1 The singular control problem

We consider the following optimal control problem in which we allow both ordinary control and singular control to act at the same time, i.e., the dynamics according to

$$x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + v_t^1 - v_t^2, \quad (s, x) \in \Sigma, \quad s \leq t \leq T,$$

We say that (v^+, v^-) is the minimal decomposition of the process v of bounded variation if v^+ and v^- are the positive and negative variations of v respectively. $v^\pm \in \mathcal{D}^d[0, T]$, $v = v^+ - v^-$ and $\tilde{v} = v^+ + v^-$, where $\tilde{v}$ is the total variation of v on $[0, T]$.

PROPOSITION 2.2 For each $\alpha = (\Omega, \mathcal{F}, \mathcal{F}_t, P, x_t, u_t, v_t^1, v_t^2, s, x) \in \Lambda_{s,x}$, we have $\bar{\alpha} = \{\mathcal{F}_t\}, P, x_t, u_t, v_t^+, v_t^-, s, x) \in \Lambda_{s,x}$ and

$$J(\bar{\alpha}) \leq J(\alpha),$$

where (v^+, v^-) is the minimal decomposition for the process $v = v^1 - v^2$. Moreover, if α is optimal, the $v^1 = v^+$, $v^2 = v^-$.

Proof The first part of the proposition is obvious, and the second part follows from the strict positivity of $c(\cdot)$ and $\bar{c}(\cdot)$.

From Proposition 2.2 we know that if the optimal control for the problem (1) then we can always find an optimal control α^* such that the two increasing (componentwise) processes v^1 and v^2 are mutually orthogonal, i.e., $dv^1 \cdot dv^2 = 0$.

2.2 The classical control problem

Following Martins and Kushner [24] (see also Zhu [30]) we now define a d -dimensional control problem which uses only the classical controls. The state problem evolves according to

$$\begin{aligned} \hat{x}^0(t) &= s + \int_0^t \hat{u}^0(\theta) d\theta, \\ \hat{x}(t) &= x + \int_0^t [b(\hat{x}^0(\theta), \hat{x}(\theta), \hat{u}(\theta)) \hat{u}^0(\theta) + \hat{n}(\theta)(1 - \hat{u}^0(\theta))] d\theta \\ &\quad + \int_0^t \sigma(\hat{x}^0(\theta), \hat{x}(\theta), \hat{u}(\theta)) \sqrt{\hat{u}^0(\theta)} d\hat{B}_\theta, \quad t > 0, (s, x) \in \Sigma. \end{aligned}$$

The control variable of this problem has the following form:

$$\hat{\mu} = (\hat{u}^0, \hat{u}, \hat{n}^1, \hat{n}^2),$$

where

- $\hat{u}^0: \mathbb{R}_+ \mapsto I = [0, 1]$,
- $\hat{u}: \mathbb{R}_+ \mapsto U$,
- $\hat{\nu} = (\hat{n}^1, \hat{n}^2): \mathbb{R}_+ \mapsto S = \{z \in \mathbb{R}_+^{2d}, \sum_{i=1}^{2d} z_i = 1\}$, and $\hat{n} = \hat{n}^1 - \hat{n}^2$.

Notice that the control set $\Xi = I \times U \times S$ is compact. Control problems of this type are discussed in Haussmann and Lepeltier [13]. Using the notation of Definition 1.1 of [13], we denote the control of this problem by

$$\hat{\alpha} = (\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathcal{F}}_t, \hat{P}, (\hat{x}_t^0, \hat{x}_t), \hat{\mu}_t, s, x).$$

$$\hat{K}(t, x) = \left\{ \left(\hat{a}(t, x, u)u^0, \hat{b}(t, x, u)u^0 + \left[\begin{pmatrix} 0 \\ n^1 \end{pmatrix} - \begin{pmatrix} 0 \\ n^2 \end{pmatrix} \right] (1 - u^0), z \right) : \right. \\ \left. z \geq f(t, x, u)u^0 + [c(t) \cdot n^1 + \bar{c}(t) \cdot n^2](1 - u^0), \quad \mu = (u^0, u, v) \in \Xi \right\},$$

where

$$\hat{a}(t, x, u) = \begin{pmatrix} 0 & 0 \\ 0 & a(t, x, u) \end{pmatrix} \in \mathcal{S}^{d+1},$$

$$\hat{b}(t, x, u) = \begin{pmatrix} 1 \\ b(t, x, u) \end{pmatrix} \in \mathbb{R}^{d+1}.$$

Note that $\hat{K}(t, x)$ is a subset of $\mathcal{S}^{d+1} \times \mathbb{R}^{d+1} \times \mathbb{R}$. It is convex according to the result since $K(t, x)$ is convex.

LEMMA 2.4 *If $K(t, x)$ is convex, then $\hat{K}(t, x)$ is also a convex set.*

Proof It is clear that we need only to show the set

$$\bar{K}(t, x) = \left\{ (a(t, x, u)u^0, b(t, x, u)u^0 + (n^1 - n^2)(1 - u^0), z) : \right.$$

$$\left. z \geq f(t, x, u)u^0 + [c(t) \cdot n^1 + \bar{c}(t) \cdot n^2](1 - u^0), \quad \mu = (u^0, u, v) \in \Xi \right\}$$

is a convex subset of $\mathcal{S}^d \times \mathbb{R}^d \times \mathbb{R}$. Let $\mu_i = (u_i^0, u_i, v_i) \in \Xi$, and

$$z_i \geq f(t, x, u_i)u_i^0 + (c(t) \cdot n_i^1 + \bar{c}(t) \cdot n_i^2)(1 - u_i^0), \quad i = 1, 2.$$

For $0 < \lambda < 1$, we want to show that $\bar{\mu} = \lambda\mu_1 + (1 - \lambda)\mu_2$ gives rise to a point in $\bar{K}$. The result is obvious if $u_1^0 = u_2^0 = 0$, so we assume $u_1^0 + u_2^0 > 0$. Let $\lambda' = \lambda u_1^0 / (u_1^0 + u_2^0)$ that $0 < \lambda' \leq 1$, and $0 \leq \lambda' \leq 1$. From the assumption that $K(t, x)$ is convex that there exists a $\bar{u} \in U$ such that

$$\lambda' a(t, x, u_1) + (1 - \lambda') a(t, x, u_2) = a(t, x, \bar{u}),$$

$$\lambda' b(t, x, u_1) + (1 - \lambda') b(t, x, u_2) = b(t, x, \bar{u}),$$

$$\lambda' f(t, x, u_1) + (1 - \lambda') f(t, x, u_2) \geq f(t, x, \bar{u}).$$

Take a control $\alpha = (\Omega, \mathcal{F}, \mathcal{F}_t, P, x(t), u(t), v^1(t), v^2(t), s, x) \in \Lambda_{s,x}$. By definition, t is a d -dimensional Brownian motion B on $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ such that

$$x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + v_t^1 - v_t^2.$$

We may assume $\mathcal{F}_t$ to be right continuous. In fact, note that B is still a Brownian motion on $(\Omega, \mathcal{F}, \mathcal{F}_{t+}, P)$, where $\mathcal{F}_{t+} = \bigcap_{t < \theta \leq T} \mathcal{F}_\theta$. Therefore $\alpha' \equiv (\Omega, \mathcal{F}, \mathcal{F}_{t+}, P, x(t), u(t), v^1(t), v^2(t), s, x) \in \Lambda_{s,x}$, with $J(\alpha') = J(\alpha)$.

In view of Proposition 2.2, we may assume that v^1 and v^2 are mutually orthogonal in other words, let

$$v(t) = v^1(t) - v^2(t),$$

we have

$$\tilde{v}_k(t) = v_k^1(t) + v_k^2(t), \quad k = 1, 2, \dots, d.$$

Let

$$\tilde{v}(t) = \sum_{k=1}^d \tilde{v}_k(t),$$

then it is obvious that

$$v_k^i(\cdot) \ll \tilde{v}(\cdot), \quad i = 1, 2, \quad k = 1, \dots, d.$$

Apply the Radon-Nikodym theorem (cf. Dellacherie and Meyer [5], Theorem 1.1). There exist processes n_k^i , $i = 1, 2$, $k = 1, \dots, d$ on $[0, T]$ which are progressively measurable with respect to $\mathcal{F}_t$ and such that

$$v_k^i(t) = \int_s^t n_k^i(\theta) d\tilde{v}(\theta), \quad i = 1, 2, \quad k = 1, \dots, d.$$

It can be easily verified that

$$\sum_{i=1}^2 \sum_{k=1}^d n_k^i(\theta) = 1$$

a.e. $(d\tilde{v}_\theta)$ on $[s, T]$. Let $n^i = (n_1^i, \dots, n_d^i)$, $i = 1, 2$. We may redefine n^i on a $d\tilde{v}_t dP$ -null set such that (3.2) holds everywhere.

The cost function of the problem can be written as

$$J(\alpha) = E^P \left\{ \int_s^T f(\theta, x_\theta, u_\theta) d\theta + \int_s^T (c(\theta) \cdot n^1(\theta) + \bar{c}(\theta) \cdot n^2(\theta)) d\tilde{v}(\theta) \right\}.$$

Let

$$\hat{M}_t = M_{T(t)}, \quad \bar{\mathcal{F}}_t = \mathcal{F}_{T(t)},$$

then the optional sampling theorem implies that $\bar{M}(t)$ is a continuous $\bar{\mathcal{F}}_t$ -martingale

$$\begin{aligned} \langle \hat{M} \rangle_t &= \int_s^{T(t)} a(\theta, x(\theta), u(\theta)) d\theta \\ &= \int_0^t a(T(\theta), x(T(\theta)), u(T(\theta))) T'(\theta) d\theta. \end{aligned}$$

Therefore, there exists a standard extension $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathcal{F}}_t, \hat{P})$ of $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$, d -dimensional Brownian motion $\hat{B}$ on this probability base such that

$$\hat{M}_t = \int_0^t \sigma(T(\theta), x(T(\theta)), u(T(\theta))) \sqrt{T'(\theta)} d\hat{B}_\theta.$$

cf. Karatzas and Shreve [17]. Define

$$\hat{\theta}^i(t) = \int_0^t n^i(T(\theta))(1 - T'(\theta)) d\theta, \quad i = 1, 2.$$

Since $\tau \circ T(t) = \min\{\theta, T(\theta) = T(t)\}$, it can be verified easily that

$$v^i(T(t)) = \hat{\theta}^i(t), \quad t \in \mathbb{R}_+ \setminus \left(\bigcup_{s \leq \theta \leq T} [(\tau(\theta), \tau(\theta +))] \right), \quad i = 1, 2,$$

and for $s \leq \theta \leq T, i = 1, 2$,

$$\hat{\theta}^i(\theta') = v^i(\theta) + n^i(\tau(\theta))(\theta' - \tau(\theta)), \quad \tau(\theta) \leq \theta' \leq \tau(\theta +).$$

Define

$$\hat{x}(t) = \begin{cases} x(T(t)), & t \in \mathbb{R}_+ \setminus \left(\bigcup_{s \leq \theta \leq T} [\tau(\theta), \tau(\theta +)] \right), \\ x(\theta) + n(\tau(\theta))(t - \tau(\theta)), & \tau(\theta) \leq t \leq \tau(\theta +), \quad s \leq \theta \leq T, \end{cases}$$

with $n = n^1 - n^2$, and

$$\hat{u}(t) = u(T(t)), \quad 0 \leq t \leq \infty.$$

Note that $T'(t) = 0$ a.e. for $t \in [\tau(\theta), \tau(\theta +)], s \leq \theta \leq T$, so we have

$$\begin{aligned} D_{T(t)} &= \int_0^t b(T(\theta), \hat{x}(\theta), \hat{u}(\theta)) T'(\theta) d\theta, \\ \hat{M}_t &= \int_0^t \sigma(T(\theta), \hat{x}(\theta), \hat{u}(\theta)) \sqrt{T'(\theta)} d\hat{B}_\theta, \quad 0 \leq t \leq \infty. \end{aligned}$$

If v^1, v^2 are not mutually orthogonal, then by Lemma 2.2 there exists a control $\tilde{\alpha} = (\Omega, \mathcal{F}, \mathcal{F}_t, P, x(t), u(t), \bar{v}^1(t), \bar{v}^2(t), s, x) \in \Lambda_{s,x}$ such that $\bar{v}^1, \bar{v}^2$ are mutually orthogonal and $J(\tilde{\alpha}) \leq J(\alpha)$. Therefore by what we have shown, there exists a control $\hat{\alpha}$ such that $\hat{J}(\hat{\alpha}) = J(\tilde{\alpha})$, and hence $\hat{J}(\hat{\alpha}) \leq J(\alpha)$.

A converse to Proposition 3.2 is the following.

PROPOSITION 3.3 *For any admissible control*

$$\hat{\alpha} = (\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathcal{F}}_t, \hat{P}, (\hat{x}^0(t), \hat{x}(t)), (\hat{u}^0(t), \hat{u}(t), \hat{n}^1(t), \hat{n}^2(t)), s, x) \in \hat{\Lambda}_{s,x},$$

there exists an admissible control $\alpha \in \Lambda_{s,x}$, which is given below by (3.12) and such that

$$J(\alpha) = \hat{J}(\hat{\alpha}).$$

Proof Define

$$\tau(t) = \inf \{ \theta : \hat{x}^0(\theta) \geq t \},$$

$$\hat{D}_t = \int_0^t b(\hat{x}^0(\theta), \hat{x}(\theta), \hat{u}(\theta)) \hat{u}^0(\theta) d\theta,$$

$$\hat{M}_t = \int_0^t \sigma(\hat{x}^0(\theta), \hat{x}(\theta), \hat{u}(\theta)) \sqrt{\hat{u}^0(\theta)} d\hat{B}(\theta).$$

Note that $\hat{u}^0(\cdot)$ is nonnegative, and therefore $\hat{x}^0(\theta)$ is nondecreasing. It can be easily seen that $\tau(\cdot)$ is nondecreasing, left continuous, $\tau(s) = 0$ and $\hat{x}^0 \circ \tau(t) = t$. Moreover, for a fixed t , $\tau(t)$ is an $\{\hat{\mathcal{F}}_\theta\}$ -stopping time since $\hat{x}^0(\cdot)$ is continuous. Define $s \leq t \leq T$,

$$\begin{cases} x(t) = \hat{x}(\tau(t)), \\ u(t) = \hat{u}(\tau(t)), \\ v^i(t) = \hat{V}^i(\tau(t)), \quad i = 1, 2, \end{cases}$$

with

$$V^i(\lambda) = \int_0^\lambda \hat{n}^i(\theta) (1 - \hat{u}^0(\theta)) d\theta.$$

By Lemma 2.3 we know that $x(\cdot), u(\cdot), v^i(\cdot)$, $i = 1, 2$, are well defined. Since $(\hat{n}^1(t), \hat{n}^2(t)) \in S$, $0 \leq \hat{u}^0 \leq 1$, and $\tau(\cdot)$ is nondecreasing and left continuous, we conclude that $v^i \in \mathcal{D}^d[0, T]$, $i = 1, 2$. Moreover, we have

$$x(t) = x + \hat{D}_{\tau(t)} + \hat{M}_{\tau(t)} + V_{\tau(t)}^1 - V_{\tau(t)}^2.$$

Evidently we now have

COROLLARY 3.4 For $(t, x) \in \Sigma$,

$$W(t, x) = \hat{W}(t, x).$$

4. EXISTENCE OF OPTIMAL CONTROL LAWS

A control

$$\alpha = (\Omega, \mathcal{F}, \mathcal{F}_P, P, x(t), u(t), v^1(t), v^2(t), s, x) \in \Lambda_{s,x}$$

is called a *control law* if there exists a Borel measurable function $\hat{u}: \Sigma \rightarrow U$ such that $u(t, \omega) = \hat{u}(t, x(t, \omega))$, and for any $s \leq t_1 \leq t_2 \leq T$,

$$v^i(t_2) - v^i(t_1) \in \sigma(x(\theta), t_1 \leq \theta \leq t_2), \quad i = 1, 2.$$

Now we can state the main theorem of the paper.

THEOREM 4.1 There exists an optimal control law for the problem (2.1).

Proof By Lemma 2.4 we know that $\hat{K}(t, x)$ is convex for any $(t, x) \in \Sigma$. Applying Theorem 2.5, there exists an optimal Markovian control

$$\hat{\alpha} = (\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathcal{F}}_P, \hat{P}, \hat{x}(t), \hat{u}(t), \hat{v}(t), s, x)$$

for the control problem (2.4), with $\hat{\mu}(t) = (\hat{u}^0(t), \hat{u}(t), \hat{v}(t))$, and

$$\hat{u}^0(t) = \hat{U}^0(\hat{x}^0(t), \hat{x}(t)), \quad \hat{u}(t) = \hat{U}(\hat{x}^0(t), \hat{x}(t)),$$

$$\hat{v}(t) = (\hat{N}^1(\hat{x}^0(t), \hat{x}(t)), \hat{N}^2(\hat{x}^0(t), \hat{x}(t))),$$

where $\hat{U}^0(\cdot, \cdot): \mathbb{R}_+ \times \mathbb{R}^d \rightarrow [0, 1]$, $\hat{U}(\cdot, \cdot): \mathbb{R}_+ \times \mathbb{R}^d \rightarrow U$, and $\hat{N}(\cdot, \cdot) = (\hat{N}^1(\cdot, \cdot), \hat{N}^2(\cdot, \cdot)): \mathbb{R}_+ \times \mathbb{R}^d \rightarrow S$ are deterministic Borel measurable functions. As we have seen in the proof of Proposition 3.3, through an extension of the filtered probability space $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathcal{F}}_P, \hat{P})$ and transforms (3.15) we can get an admissible control

$$\alpha = (\Omega, \mathcal{F}, \mathcal{F}_P, P, x(t), u(t), v^1(t), v^2(t), s, x).$$

By Proposition 3.3 and Corollary 3.4, we know that α is optimal, i.e., $J(\alpha) = J^*$. It remains to show that α is a control law. Note that v^1, v^2 are orthogonal to each other by Proposition 2.2. Since $\hat{x}^0(\cdot)$ is continuous, we have $\hat{x}^0(\tau(t)) = t$ for $s \leq t \leq T$. Therefore

$$u(t) = \hat{u}(\tau(t)) = \hat{U}(\hat{x}^0(\tau(t)), \hat{x}(\tau(t))) = \hat{U}(t, x(t)).$$

To show (4.1), let

$$J \equiv \{t: x(t) \neq x(t+), s \leq t \leq T\}.$$

where $(\Delta x_\theta)^+ = \max\{x(\theta+) - x(\theta), 0\}$, $(\Delta x_\theta)^- = -\min\{x(\theta+) - x(\theta), 0\}$. The

$$H(t) = \sum_{s \leq \theta < t} \Delta x_\theta,$$

where $\Delta x_\theta = x(\theta+) - x(\theta)$.

We consider the process $I^i(\cdot)$. By Lemma A.2, and (4.4), noticing that $\tau(t) > \theta$ only if $\hat{x}^0(\theta) < t$, we can write

$$\begin{aligned} I^i(t) &= \int_0^{\tau(t)} \mathbf{1}_{\{\hat{x}^0(\theta) \notin J\}}(\theta) \hat{N}^i(\hat{x}^0(\theta), \hat{x}(\theta)) (1 - \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\theta))) d\theta \\ &= \int \mathbf{1}_{\{0 \leq (\theta) < \tau(t)\}}(\theta) \mathbf{1}_{\{\hat{x}^0(\theta) \notin J\}}(\theta) \hat{N}^i(\hat{x}^0(\theta), \hat{x}(\theta)) (1 - \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\theta))) d\theta \\ &= \int \mathbf{1}_{\{s \leq \hat{x}^0(\theta) < t\}}(\theta) \mathbf{1}_{\{\hat{x}^0(\theta) \notin J\}}(\theta) \hat{N}^i(\hat{x}^0(\theta), \hat{x}(\theta)) (1 - \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\theta))) d\theta \\ &= \int \mathbf{1}_{\{s \leq \hat{x}^0(\theta) < t; \hat{x}^0(\theta) \notin J\}}(\theta) \hat{N}^i(\hat{x}^0(\theta), \hat{x}(\tau(\hat{x}^0(\theta)))) (1 - \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\tau(\hat{x}^0(\theta)))) d\theta \\ &= \int \mathbf{1}_{\{s \leq \theta < t, \theta \notin J\}}(\theta) \hat{N}^i(\theta, \hat{x}(\tau(\theta))) (1 - \hat{U}^0(\theta, \hat{x}(\tau(\theta)))) d\tau(\theta) \\ &= \int_s^t \mathbf{1}_{J^c}(\theta) \hat{N}^i(\theta, x(\theta)) (1 - \hat{U}^0(\theta, x(\theta))) d\tau(\theta) \\ &= \int_s^t \hat{N}^i(\theta, x(\theta)) (1 - \hat{U}^0(\theta, x(\theta))) d\tau^c(\theta), \end{aligned}$$

where $\tau^c(\cdot)$ denotes the continuous part of the increasing process $\tau(\cdot)$. Define

$$R \equiv \{\theta: \hat{U}^0(\theta, x(\theta)) = 0, \quad s \leq \theta \leq T\}.$$

From the definition $\tau(\cdot)$, we know

$$t = s + \int_0^{\tau(t)} \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\theta)) d\theta, \quad s \leq t \leq T.$$

As above, we can get

$$t - s = \int_0^t \hat{U}^0(\theta, x(\theta)) d\tau^c(\theta), \quad s \leq t \leq T,$$

since $\hat{u}^0(\theta) = 0$ if $\hat{x}^0(\theta) \in J$ by (4.5). Therefore for any nonnegative Borel function defined on $[s, T]$,

$$\int_{[s, T] \cap R^c} k(\theta) d\tau^c(\theta) = \int_{[s, T] \cap R^c} \frac{1}{\hat{U}^0(\theta, x(\theta))} k(\theta) d\theta.$$

Let $v = v^1 - v^2$, we have shown that

$$\begin{aligned} v(t) &= I(t) + II(t) = I^{ac}(t) + I^{sc}(t) + II(t) \\ &= \int_s^t (\hat{N}^1(\theta, x(\theta)) - \hat{N}^2(\theta, x(\theta))) \frac{1 - \hat{U}^0(\theta, x(\theta))}{\hat{U}^0(\theta, x(\theta))} \mathbf{1}_{\{\hat{U}^0(\theta, x(\theta)) \neq 0\}}(\theta) d\theta \\ &\quad + \int_s^t \mathbf{1}_{\{\theta: \hat{U}^0(\theta, x(\theta)) = 0\}} dx^c(\theta) + \sum_{s \leq \theta < t} \Delta x_\theta, \end{aligned}$$

and the three parts are the absolute continuous, singular continuous and jump of the bounded variation process $v(\cdot)$ respectively. (4.1) is obviously satisfied by

Since v^1, v^2 are orthogonal to each other, we have $\tilde{v} = v^1 + v^2$. From the def of total variation we can conclude that (4.1) is satisfied by $\tilde{v}$. The proof of the th is therefore completed by noticing that $v^1 = \frac{1}{2}(\tilde{v} + v)$, $v^2 = \frac{1}{2}(\tilde{v} - v)$.

5. SOME COMMENTS

(a) The method we have used can be applied to the following more general problem which is similar to the one formulated in Fleming and Soner [9], Chapter 8. Let

$$B = \{n \in \mathbb{R}^d: \|n\| \leq 1\}, \quad S = \{n \in \mathbb{R}^d: \|n\| = 1\}$$

and let C be a closed cone in $\mathbb{R}^d$, i.e.,

$$n \in C, \quad \lambda > 0 \Rightarrow \lambda n \in C.$$

Define $S^0 = S \cap C$. Assume that $c(\cdot, \cdot): [0, T] \times C \mapsto \mathbb{R}_+^d$ is continuous and $c > 0$. We also assume that $c(\cdot, \cdot)$ satisfies the following condition

$$c(t, \lambda x) = \lambda c(t, x)$$

for $0 \leq \lambda \leq 1$, $x \in B \cap C$. Instead of the controls v^1, v^2 as in (2.1), we use

$$v(t) = \int_0^t n(\theta) d\xi(\theta).$$

The state equation thus becomes

$$x_t = x + \int_s^t b(\theta, x_\theta, u_\theta) d\theta + \int_s^t \sigma(\theta, x_\theta, u_\theta) dB_\theta + \int_s^t n(\theta) d\xi(\theta), \quad (s, x) \in \Sigma, \quad s \leq t$$

and the singular control is expressed by a pair of processes $n(\cdot): [0, T] \mapsto S^0$, $\xi: [0, T] \mapsto \mathbb{R}_+$, progressively measurable with respect to $\mathcal{F}_t$, with $\xi(s) = 0$ and

from n . Define

$$n(\theta) = \begin{cases} \bar{N}(\theta, \hat{x}_{\tau(\theta)}), & \text{if } \tau(\theta) = \tau(\theta +), \\ \frac{\hat{x}(\tau(\theta +)) - \hat{x}(\tau(\theta))}{\tau(\theta +) - \tau(\theta)}, & \text{otherwise.} \end{cases}$$

Then $n(\cdot)$ is progressively measurable with respect to $\mathcal{F}_t$. We also define

$$\xi(t) = \int_0^{\tau(t)} \|\hat{N}(\hat{x}^0(\theta), \hat{x}(\theta))\| (1 - \hat{U}^0(\hat{x}^0(\theta), \hat{x}(\theta))) d\theta.$$

The proof for the property (4.1) is similar for

$$v(t) \equiv \int_0^t n(\theta) d\xi(\theta).$$

Unfortunately, we are unable to show that there exists a pair $(N(\cdot, \cdot), \xi(\cdot))$ such that $n(t) = N(t, x(t))$ and ξ satisfies (4.1).

(b) It is well known that under some conditions the value function $W(\cdot)$ of the classical control problem in Section 2.2 satisfies the following Hamilton–Jacobi–Bellman equation

$$\inf_{\mu \in \Xi} \{ \hat{\mathcal{L}}^\mu W(t, x) + \hat{f}^\mu(t, x) \} = 0$$

in some generalized sense (e.g., in the viscosity sense), cf. Krylov [19], Lions [20], where $\mu = (u^0, u, n^1, n^2) \in \Xi$, and

$$\hat{\mathcal{L}}^\mu = \sum_{i,j=0}^d \hat{a}_{ij}^\mu(t, x) \frac{\partial^2}{\partial \hat{x}^i \partial \hat{x}^j} + \sum_{i=0}^d \hat{b}_i^\mu(t, x) \frac{\partial}{\partial \hat{x}^i},$$

$$\hat{f}^\mu(t, x) = f(t, x, u)u^0 + [c(t) \cdot n^1 + \bar{c}(t) \cdot n^2](1 - u^0)$$

with $\hat{x}^0 = t$, $\hat{x}^i = x^i$, $1 \leq i \leq d$, and

$$\hat{a}^\mu(t, x) = \begin{pmatrix} 0 & 0 \\ 0 & a(t, x, u)u^0 \end{pmatrix},$$

$$\hat{b}^\mu(t, x) = \begin{pmatrix} u^0 \\ b(t, x, u)u^0 + (n^1(t) - n^2(t))(1 - u^0) \end{pmatrix}.$$

We can rewrite (5.1) as

$$\inf_{\mu \in \Xi} \{ u^0 [\mathcal{L}^\mu W(t, x) + f(t, x, u)] + (1 - u^0) [n^1 \cdot (\nabla_x W(t, x) + c(t)) + n^2 \cdot (-\nabla_x W(t, x) + \bar{c}(t))] \} =$$

Let $u^0(\cdot)$, $k(\cdot)$ be nonnegative Borel measurable functions on $\mathbb{R}_+$, and $0 \leq u^0$. Define

$$f(t) = s + \int_0^t u^0(\theta) d\theta, \quad 0 \leq t < \infty,$$

$$g(t) = \inf \{ \theta : f(\theta) \geq t \} \quad (\inf \emptyset = \infty),$$

$$v(t) = \int_0^{g(t)} k(\theta) d\theta.$$

The following results are used in the proofs of Proposition 3.3 and Theorem

LEMMA A.2 (a) For any nonnegative Borel measurable function $l(\cdot)$ on $[s, T]$,

$$\int_0^{g(T)} l(f(\theta)) d\theta = \int_s^T l(\theta) dg(\theta).$$

(b) For any nonnegative Borel measurable function $\bar{l}(\cdot)$ defined $[s, T]$,

$$\int_0^T \bar{l}(\theta) dv(\theta) = \int_0^{g(T)} \bar{l}(f(\theta)) k(\theta) d\theta.$$

Proof (a) As in the proof of Lemma A.1, we need only to show the case where $\mathbf{1}_{[s,t]}(\theta)$ for any $t \in [0, T]$. Note that $g(t) > \theta$ if and only if $f(\theta) < t$, therefore

$$\int_0^{g(T)} \mathbf{1}_{[s,t]}(f(\theta)) d\theta = m(\theta : s \leq f(\theta) < t) = m(\theta : 0 \leq \theta < g(t)) = g(t),$$

which is exactly the right hand side of (A.2) (m is the Lebesgue measure on $\mathbb{R}$).

(b) As before, we may assume $\bar{l}(\theta) = \mathbf{1}_{[s,t]}(\theta)$ for some $s \leq t \leq T$. Then it that

$$\begin{aligned} \int_0^{g(T)} \mathbf{1}_{[s \leq f(\theta) < t]}(\theta) k(\theta) d\theta &= \int_0^{g(T)} \mathbf{1}_{\{\theta < g(t)\}}(\theta) k(\theta) d\theta \\ &= \int_0^{g(t)} k(\theta) d\theta = v(t), \end{aligned}$$

which, by definition, is equal to the left hand side of (A.3).

Remark A.3 Dellacherie and Meyer [5] VI 54–55 gives a slightly different more general form of Lemma A.2(a).